

Backtest of ICES Advice for Selected Northeast Atlantic Stocks

Technical report

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Abstract

[TODO: 2–3 paragraph abstract summarising aims, case studies, methods (perfect-management peels), and headline findings once backtests are complete.]

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1 Introduction

[TODO: Motivation: need to separate assessment/advice robustness from implementation of advice; policy relevance for HCRs and rebuilding.]

This report documents a backtest of fisheries advice for selected Northeast Atlantic stocks. Stock assessments are reconstructed annually over a historical period (retrospective peels). The advice rule in operation each year is applied, and stocks are projected forward under perfect implementation of catch advice. Counterfactual outcomes are compared with realised catches, stock status, and risks associated with overfishing.

1.1 Aims and objectives

1. **Reconstruct** the annual perception of stock status using only data and methods available at the time.
2. **Reproduce** the harvest control rule (ICES advice rule) in force each year.
3. **Simulate** forward under perfect management (catches match advised catch).
4. **Compare** counterfactual outcomes with realised history.

1.2 Report structure

Section 2 describes the backtest procedure, including operating-model conditioning (Section 2.3). Section 3 lists confirmed case-study stocks. Section 4 summarises week 1 screening and benchmarks. Section 5 presents stock-specific results. Section 6 provides a cross-stock synthesis. Section 7 discusses implications and limitations. Section 9 documents software and reproducibility.

2 Methods: backtest procedure

The study separates two questions that are often conflated: whether the assessment and advice rule are robust, and whether management outcomes differ because advice was not followed. The closed-loop procedure (perceive–advise–implement–compare) follows the **FLBacktest** Methods vignette. Historical outcomes may differ from those expected under advice not only because recommended catches were not implemented exactly, but also because stock perception, reference points, and recruitment changed through time. The backtest separates these effects for the six Northeast Atlantic ICES Category 1 stocks in Table 1.

2.1 Historical assessment reconstruction

[TODO: Detail peel configurations (SAM / equivalent), data cuts, validation against published ICES assessments, and any shortcuts used.]

For each stock and peel year t , the assessment is re-run or reconstructed using only contemporaneous information. The peel supplies the perception of SSB, F , and recruitment used in the advice step—not the latest revised history. The effective window is about 7–10 assessment years (roughly 2016–2026), set by SAG online coverage. Short-cut runs may replace full peels with multiplicative SSB error conditioned on retrospective bias (**retroBias**, **shortcut**). This is a simulation experiment, not a statutory re-assessment.

2.2 Advice rule

- **Celtic and Irish Sea gadoids:** ICES Category 1 MSY approach— $F = F_{\text{MSY}}$ above $\text{MSY}B_{\text{trigger}}$, reduced linearly towards B_{lim} , and zero at or below B_{lim} .
- **Northeast Atlantic mackerel:** Coastal States / ICES MSY rule (including any stability constraints applying in year t).

Reference points are those published or used for advice in year t . Benchmark years are treated as structural breaks when F_{MSY} , B_{lim} , or $\text{MSY}B_{\text{trigger}}$ are redefined.

2.3 Operating-model conditioning

The OM representing *true* stock dynamics for each case study is taken from the most recent accepted age-based assessment, stored as a local `FLStock`. The six OMs cover Celtic Sea cod, whiting, and haddock; Irish Sea cod and whiting; and Northeast Atlantic mackerel. Conditioning prepares each assessment for closed-loop projection under `FLBacktest::hcrICES` and `FLasher`. The critical biological step is the *stock–recruitment relationship*: without an assessment-consistent relationship and recruitment deviates, short-term counterfactuals conflate fishing-policy effects with year-class strength (Section 4).

Data. Each OM uses the windowed `FLStock` (SSB, F , recruitment, catch, biology, and selectivity) from `data/reference/stocks.csv`; SAG assessment-year series and ICES advice over the peel window; and HCR parameters F_{MSY} , B_{lim} , and $\text{MSY}B_{\text{trigger}}$ from SAG (FLBRP or recent assessment quantities only as documented fallbacks).

Procedure. For each stock:

1. Load and window the `FLStock` to the advice year; clean discard and landings slots so that `FLasher` projections are well defined.
2. Fit a Beverton–Holt stock–recruitment relationship with *dynamic virgin biomass* (B_0) via `ftmb2` $\rightarrow$ `b0dyn` $\rightarrow$ `ftmb_b0dyn` (steepness prior $s = 0.7$). Unfished biomass can vary with historical biology and selectivity. Retain the fitted relationship and log-scale recruitment deviates.
3. Build an FLBRP equilibrium object and attach HCR parameters `ftar`, `blim`, and `btrig` from SAG. An implausible FLBRP F_{MSY} is not substituted for the SAG advice parameter.

Conditioned OMs are written to `data/results/om.RData` (`Rmd/02_condition_om.Rmd`).

Recruitment treatments. Screening used geometric-mean recruitment as a simple proxy to show how strongly trajectories depend on year-class strength; that proxy is not the recruitment model for the full backtest. Counterfactual runs use the fitted stock–recruitment relationship and propagate recruitment deviates. Deterministic mean-recruitment runs are retained only as a sensitivity comparison, so that fishing mortality and recruitment can be separated: comparing mean vs deviate paths holds policy fixed and isolates year-class effects; comparing deviate-informed paths with the realised assessment isolates departures of realised catch (or F) from advice.

Case-study caveats. Northeast Atlantic mackerel requires care at the recent benchmark (scale and reference-point revision); results remain provisional until benchmark-year handling is agreed. Irish Sea whiting has sparse recent SAG coverage and was windowed to an earlier terminal year; it remains in the set with peel diagnostics flagged.

SAG availability limits the stock-specific retrospective peel windows to approximately 7–10 assessment years. This is a simulation study, not a statutory re-assessment of any of the six stocks.

2.4 Perfect-management projection

Closed-loop simulation uses `FLBacktest::hcrICES` with `FLasher` projections. From each peel (or short-cut perception):

1. Reconstruct the stock status perceived from information available in year t (not the latest revised estimate).
2. Apply the contemporaneous HCR and its reference points to obtain advised catch for the next management year.
3. Set realised catch equal to advised catch (perfect implementation: no TAC overshoot, override, or implementation error) and project the conditioned OM forward under the chosen recruitment treatment.

Conceptual contrast. Over the backtest window we compare:

1. **Realised trajectory:** history implied by the conditioned OM when forced by observed catch (or realised F), with assessed recruitment deviates.
2. **Counterfactual trajectory:** history under perfect implementation of contemporaneous advice in the same OM.

Both use the same conditioned biological dynamics and stock–recruitment relationship. Differences describe the effect of the counterfactual fishing history under the stated OM assumptions; they are not evidence that a new assessment should replace the accepted ICES assessment. The short peel window and OM assumptions bound the inferences.

2.5 Evaluation metrics

- **Biological status** — time below B_{lim} and $MSY B_{\text{trigger}}$; rebuilding times to $MSY B_{\text{trigger}}$ or another specified threshold;
- **Exploitation** — realised and counterfactual F relative to F_{MSY} ;
- **Yield and stability** — annual and cumulative catch, and interannual variability.

3 Case-study stocks

[TODO: Replace with generated/week1_tab_stocks.tex or an updated table once the list is final.]

All six proposal candidates were retained after week 1 screening (see Section 4).

Table 1: Confirmed case-study stocks.

Stock	SID	ICES area	Advice basis
Celtic Sea cod	<code>cod.27.7e-k</code>	7.e-k	ICES MSY approach
Celtic Sea whiting	<code>whg.27.7b-ce-k</code>	7.b-c, 7.e-k	ICES MSY approach
Celtic Sea haddock	<code>had.27.7b-k</code>	7.b-k	ICES MSY approach
Irish Sea whiting	<code>whg.27.7a</code>	7.a	ICES MSY approach
Irish Sea cod	<code>cod.27.7a</code>	7.a	ICES MSY approach
NE Atl. mackerel	<code>mac.27.nea</code>	1-8, 14, 9.a	MSY rule (Coastal States)

4 Week 1 screening and benchmarks

[TODO: Optionally \input narrative or tables from `week1_summary.tex` / generated/. For a short report, keep a condensed summary and cite the standalone week 1 PDF.]

Week 1 confirmed case studies, screened SAG coverage and latest status, and ran historical projections at target F . Key discussion points for the full backtest:

- Perception of the stock changes at each update or benchmark.
- Recruitment has a large impact on short-term trajectories.
- Reference points are routinely changed at a benchmark.

Figure placeholder: SAG SSB retrospectives (week 1)

`../output/figures/week1_sag_ssb.pdf`

Figure 1: SAG SSB retrospectives (week 1)

Figure placeholder: SAG vs FLStock scaled trajectories (week 1)
../output/figures/week1_sag_fl.pdf

Figure 2: SAG vs FLStock scaled trajectories (week 1)

Figure placeholder: Historical projections at target F (week 1)
../output/figures/week1_projections.pdf

Figure 3: Historical projections at target F (week 1)

5 Results by stock

[TODO: One subsection per stock. Pattern below; duplicate and fill.]

5.1 Celtic Sea cod (cod.27.7e-k)

Assessment reconstruction. [TODO: Peel years, diagnostics, agreement with published assessments.]

Advice and perfect-management counterfactual. [TODO: Describe HCR application and projection settings.]

Figure placeholder: Celtic Sea cod: realised vs counterfactual SSB

../output/figures/results_cod27.7e-k_ssb.pdf

Figure 4: Celtic Sea cod: realised vs counterfactual SSB

Performance metrics. [TODO: Table of status, exploitation, and yield metrics.]

Table 2: Celtic Sea cod — illustrative metrics table (replace with results).

Metric	Realised	Counterfactual	Difference
Years $SSB < B_{lim}$	—	—	—
Years $SSB < MSY B_{trigger}$	—	—	—
Mean F/F_{MSY}	—	—	—
Mean catch	—	—	—

5.2 Celtic Sea whiting (whg.27.7b-ce-k)

[TODO: Fill following the same pattern as Section 5.1.]

5.3 Celtic Sea haddock (had.27.7b-k)

[TODO: Fill following the same pattern as Section 5.1.]

5.4 Irish Sea whiting (whg.27.7a)

[TODO: Fill following the same pattern as Section 5.1.]

5.5 Irish Sea cod (cod.27.7a)

[TODO: Fill following the same pattern as Section 5.1.]

5.6 Northeast Atlantic mackerel (mac.27.nea)

[TODO: Fill following the same pattern as Section 5.1; note Coastal States rule specifics.]

6 Cross-stock synthesis

[TODO: Where adherence would have improved outcomes; where effect was limited; where shortcomings reflect the advice rule or assessment revision rather than implementation failure.]

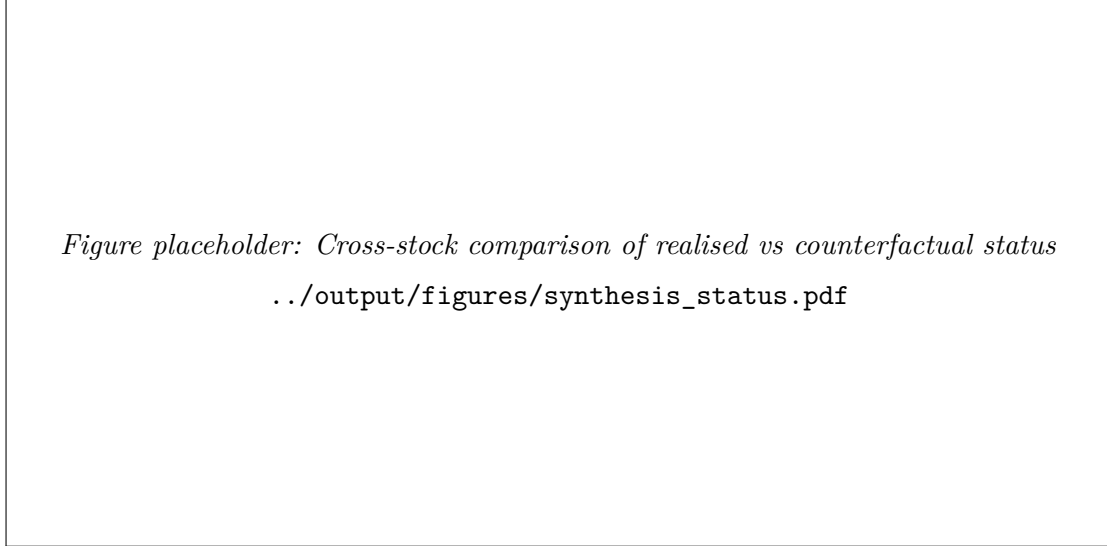


Figure 5: Cross-stock comparison of realised vs counterfactual status

Table 3: Cross-stock summary (placeholder).

Stock	Status improved?	Yield change	Main driver
Celtic Sea cod	—	—	—
Celtic Sea whiting	—	—	—
Celtic Sea haddock	—	—	—
Irish Sea whiting	—	—	—
Irish Sea cod	—	—	—
NE Atl. mackerel	—	—	—

7 Discussion

7.1 Perception, recruitment, and reference points

[TODO: Expand using week 1 discussion: (i) perception changes at updates / benchmarks; (ii) recruitment impact; (iii) reference points change at benchmarks—and how the backtest design addresses each.]

7.2 Limitations

[TODO: Assessment approximation error; advice-rule simplifications; recruitment assumptions; data gaps (e.g. Irish Sea whiting); perfect-implementation idealisation.]

7.3 Implications for advice and management

[TODO: HCR design, rebuilding, and value of following advice vs revising assessments.]

8 Conclusions

[TODO: Bullet conclusions tied to objectives; confirmed stocks; headline counterfactual findings; recommended follow-up.]

9 Software and reproducibility

Analysis uses FLR (FLCore, FLBRP, Flasher, ggplotFL) and FLBacktest / backtest (hcrICES, peel and projection helpers). Notebooks write `data/results/*.RData`; this report loads figures/tables generated from those objects.

Table 4: Key paths.

Path	Role
Rmd/01_screening.Rmd	Week 1 screening
Rmd/02_stock_summary.Rmd	SAG vs FLStock; target- F projections
Rmd/04_backtest.Rmd	Peels and counterfactuals
scripts/build_week1_tex.R	Week 1 LaTeX tables/figures
tex/week1_summary.tex	Week 1 deliverable
tex/report_template.tex	This technical report template
data/reference/stocks.csv	Stock metadata

Compile.

```
cd tex
pdflatex report_template.tex
pdflatex report_template.tex
```

Acknowledgements

[TODO: Optional.]

A Supplementary tables and figures

[TODO: Extra diagnostics, peel-by-peel plots, sensitivity runs.]